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D020811(020)

B. Tech. (Eighth Semester) Examination, April-May 2023

(AICTE Scheme)

(Civil Engineeing Branch)

STRUTURAL ENGINEERING DESIGN-IV

Time Allowed : Four hours

Maximum Marks : 100

Minimum Pass Marks : 35

Note : *Attempt all questions. Part (a) from each question is compulsory. Solve any one part from remaining parts of each question. Use of relevant IS and IRC code is permitted. Take Grade of concrete as M20 and that of steel as Fe 415 unless otherwise specified. The figure in the right hand margin indicates marks.*

Unit-I

1. (a) Briefly explain the following :

(i) Redistribution of Moments 2

(ii) Raft foundation 2

(b) Design a strap footing for two columns A and B, spaced 5 m centre to centre. Column A is on property line (300 mm × 300 mm, in size) and Column B (400 mm × 400 mm, in size) carries

service load of 400 kN and 600 kN respectively. The bearing capacity of soil is 120 kN/m². 16

(c) Design the three span continuous beam by applying moment redistribution (to the extent permitted by the code) having equal span length of 8 m subjected to total load of 30 kN/m. 16

Unit-II

2. (a) List the different types of retaining wall and briefly explain with figure counterfort retaining wall. 4

(b) Design a cantilever retaining wall structure (Toe and Heel Slab) which is required to support a 4·0 m high bank of earth above the ground level on the toe side of the wall. Consider the backfill surface to be level but subjected to a surcharge of 40 kN/m². Assume good soil for foundation at a depth of 1·25 m below the ground level with a safe bearing capacity of 160 kN/m². Take unit weight of backfill to be 16 kN/m³, angle of internal friction is 30° and coefficient of friction between soil and concrete to be 0·50. 16

(c) Design a suitable counterfort retaining wall to support a 7·50 m high bank of earth above the ground level on the toe side of the wall. Assume good soil for foundation at a depth of 1·50 m below the ground level with a safe bearing capacity of 170 kN/m². Take unit weight of backfill to be 16 kN/m³, angle

of shearing resistance of 30° and coefficient of friction between soil and concrete to be 0·50. 16

Unit-III

3. (a) Describe briefly Intze tank. 4

(b) Design a circular water tank (using approximate method of analysis) for a capacity of 400000 litres. The depth of water in the tank is to be 4m including a free board of 200 mm. Use M20 grade concrete and Fe 250 grade steel. Assume the joint between tank wall and base to be rigid. 16

(c) Design a circular water tank (using membrane analysis) with domed bottom and top for a capacity of 400000 litres including a free board of 20 cm. Take central rise of top dome as 2 m and bottom dome as 2·2 m. Assume the tank to be supported on a masonry tower. 16

Unit-IV

4. (a) List different types of loads, forces and stresses considered in the design of bridges. 4

(b) Design a solid slab bridge for class A loading for the following data :

Clear span = 4·5 m.

Clear width of roadways = 7 m.

Average thickness of wearing coat = 80 mm.

Take unit weight of concrete as 24 kN/m³ and

σ_{st} = 125 N/mm². 16

(c) Design a T-beam bridge for two lanes of class A loading for the following data:

Centre to centre span of bearings = 16 m.

Clear width of roadways = 7 m.

Average thickness of wearing coat = 80 mm.

Take unit weight of concrete as 24 kN/m³ and

σ_{st} = 125 N/mm². 16

Unit-V

5. (a) Define prestressed concrete and write minimum grade of concrete required for pre tensioned and post tensioned member. 4

(b) What are the various losses in prestressing? Explain with formulae. 16

(c) A simply supported prestressed concrete beam of rectangular cross-section 400 mm × 600 mm, is loaded with a total U.D.L. of 256 kN over a span of 6 m. Sketch the distribution of stresses at mid-span and end sections if the prestressing force is 1920 kN and the tendon is :

(i) Concentric

(ii) Eccentric, located at eccentricity of 100 mm 16